Radiotherapy for soft tissue sarcoma

Radiotherapy uses high-energy rays to destroy cancer cells, while doing as little harm as possible to normal cells. It is an important treatment for soft tissue sarcomas.

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About radiotherapy for soft tissue sarcoma

Radiotherapy uses high-energy rays to destroy cancer cells, while doing as little harm as possible to normal cells. It is an important part of treatment for soft tissue sarcomas.

Radiotherapy for sarcoma can be given:

- before surgery, to shrink the tumour and make it easier for the surgeon to remove it, or to avoid having to amputate the limb (neo-adjuvant treatment)
- after surgery, to destroy any remaining cancer cells and reduce the risk of the cancer coming back (adjuvant treatment)
- to relieve symptoms and control a sarcoma that has come back after treatment or spread to another part of the body (palliative radiotherapy)
- rarely, on its own as the main treatment.

How radiotherapy is given

There are different techniques and ways of having radiotherapy for soft tissue sarcoma.

Intensity-modulated radiation therapy (IMRT)

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This type of radiotherapy is often used to treat sarcoma. IMRT shapes the radiotherapy beams and gives the tumour a high dose of radiation. It allows different doses of radiotherapy to be given to different areas. Lower doses are given to nearby healthy tissue. This can help reduce immediate side effects and late effects.

Volumetric-modulated arc radiotherapy (VMAT)

VMAT is a newer way of giving IMRT. The radiotherapy machine moves around you and reshapes the beam during treatment. This makes it more accurate and shortens the treatment time.

Stereotactic radiotherapy

This is a new way of giving radiotherapy that may sometimes be used to treat a small soft tissue sarcoma. You have stereotactic radiotherapy treatment from a specially adapted radiotherapy machine. It gives beams of radiotherapy from many different angles. The beams overlap at the tumour. This gives the tumour a very high dose of radiotherapy. Surrounding tissues only get a very small dose.

Stereotactic radiotherapy is not available in all hospitals in the UK. Your specialist can give you more information if this is an option for you.

Proton beam therapy

Proton beam therapy uses proton radiation to destroy cancer cells, instead of x-rays. Proton beams can be made to stop when they leave the area being treated. This means it causes very little damage to nearby healthy tissue and fewer side effects. Proton beam therapy is sometimes used to treat a sarcoma very close to important structures, such as the spine.

Brachytherapy

Brachytherapy is a type of internal radiotherapy. Doctors treat the cancer by inserting radioactive material directly into the affected area. The tumour gets a high dose of radiation, but healthy tissue only gets a small amount. Brachytherapy is not commonly used to treat soft tissue sarcoma. If your specialist thinks brachytherapy will be helpful for you, they will discuss the treatment and its side effects with you.

Intraoperative radiation therapy (IORT)

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IORT is a type of internal radiotherapy. It is done during surgery. After removing the cancer, the doctor gives the same area a single dose of radiotherapy from a special machine. Some people may also have another type of radiation after surgery. IORT is not suitable for everyone and is not widely available on the NHS.

Planning your radiotherapy

Before you start radiotherapy, the radiotherapy team will plan your treatment. They make sure the radiotherapy is aimed directly at the cancer, causing the least possible damage to nearby healthy tissue. Planning may take a few visits.

Radiotherapy masks and moulds

Radiotherapy moulds may be used to help keep the part of the body having treatment still during treatment. If you are having radiotherapy to your head and neck, you will have a head and neck mould made. If you are having radiotherapy to a limb, you may also need a mould for it. Your cancer doctor or specialist nurse will tell you more about moulds if you need one.

Skin markings

To help the radiographers position you accurately for the treatment, they may make marks on your mould or mask. If you do not have a mould or mask, they will make the marks on your skin. These marks must stay throughout your treatment.

Some marks can be washed off when your course of treatment finishes. Other marks may be permanent. These are very small. They will only be made with your permission. It can be uncomfortable while the marks are being made, but only for a very short time.

As part of the planning, you will have a CT scan of the area to be treated. This helps plan the precise area for your radiotherapy. This session usually takes about 30 minutes. You may also need to have an MRI scan.

The information from the scans goes into the radiotherapy planning computer. The computer precisely designs your individual treatment plan.

Having radiotherapy for soft tissue sarcoma

Radiotherapy is normally given as a series of short, daily treatments in the hospital radiotherapy department

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The treatments are usually given every day from Monday to Friday. The number of treatments you have depends on the type, size and position of the area to be treated.

A course of radiotherapy for an early-stage cancer usually lasts about 6 weeks. Each treatment takes about 10 to 15 minutes. Your doctor will talk to you about the treatment and possible side effects.

At the start of each session of radiotherapy, the radiographer will explain what to expect. They will position you carefully on the treatment couch in the same way as during the planning session. When you are in the right position, the radiographers will leave the room and you will have treatment. Radiotherapy is not painful, but you have to lie still for a few minutes. The radiographers will be able to see and hear you during your treatment.

If you are having stereotactic radiotherapy treatment or proton beam therapy treatment, sessions are usually longer. But you will need fewer sessions than other types of radiotherapy. Your doctor, nurse or radiographer will tell you what to expect if you are having these specialised types of radiotherapy.

Support from Macmillan

Macmillan is here to support you. If you would like to talk, you can do the following:

- Call the Macmillan Support Line for free on 0808 808 00 00.
- Chat to our specialists online.
- Visit our radiotherapy and side effects forum to talk with people who have been affected by radiotherapy, share your experience, and ask an expert your questions.

Side effects of radiotherapy for soft tissue sarcoma

You may develop side effects during your treatment. These side effects will usually disappear over a few weeks or months after treatment finishes.

Your cancer doctor, nurse or radiographer will explain the ones you are most likely to get, so you know what to expect. They will also talk to you about late effects of radiotherapy. These are side effects that may not go away, or that appear months or years later.

Always tell them about any side effects you have during or after treatment. There are often things that can be done to help.

Skin changes

You may find your skin in the treatment area:

- becomes red
- darkens
- feels sore
- feels itchy.

Sometimes the skin gets very sore. It may blister, break or leak fluid. Very rarely, your doctor may stop your treatment for a short time to allow a serious skin reaction to recover.

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Skin reactions can take time to improve. They are usually better about 4 weeks after you finish treatment. In the longer term, the area of skin may look or feel slightly different to the surrounding skin.

Your radiographers or nurses will give you advice on how to look after your skin during and after treatment.

Tiredness (fatigue)

Tiredness is a common side effect and may continue for months after treatment finishes. Try to get plenty of rest but balance this with some gentle exercise, such as walking. This can improve your energy levels and help you to feel better.

After your treatment finishes, you can gradually increase your activity.

Hair loss

Radiotherapy can make your hair fall out in the area being treated. It may grow back after treatment finishes, but for some people the hair loss is permanent. This depends on how much radiotherapy you have. Your cancer doctor or radiographer can tell you if your hair is likely to grow back after treatment.

Feeling sick (nausea)

If the treatment area is near the tummy and pelvis you may feel sick and sometimes be sick (vomit). Your doctor can prescribe anti-sickness drugs to control this. These are called anti-emetics. Take them as your doctor has prescribed and let them know if a drug is not working. They can prescribe a different anti-sickness drug.

Long-term side effects of radiotherapy

Some people may have long-term or late effects of radiotherapy. These can develop months or sometimes years after treatment. Always let your cancer doctor or specialist nurse know if you have any new symptoms or problems after treatment.

Lymphoedema

After radiotherapy, some people may develop swelling called lymphoedema. It happens because the lymph nodes and vessels can be damaged by radiotherapy or surgery. Lymph fluid, which circulates around the lymphatic system, cannot pass along the vessels. It builds up, causing swelling.

There are things you can do to help reduce the risk of lymphoedema. Let your doctor know about any swelling you have after radiotherapy treatment.

Stiff joints

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Radiotherapy to a joint, such as the knee or elbow, may cause it to become stiff. Talk to your cancer doctor if you notice this. They can prescribe painkillers to help.

It is important to keep the joint mobile by using it and doing regular exercise. A physiotherapist will give you some exercises to do before, during and after treatment to help prevent stiffness.

Bone changes

Radiotherapy to a limb can increase the risk of breaking (fracturing) a bone in the area. This is rare. Most people who have radiotherapy will not have any bone problems.

If you have pain or aching in the bones, always let your cancer doctor or nurse know. It can be caused by different conditions, but it is very important to get it checked.

Second cancer

A small number of people will develop a second cancer because of their radiotherapy treatment. The chance of developing a second cancer is very small. The benefits of radiotherapy are much greater than the risks.

About our information

References

Below is a sample of the sources used in our soft tissue sarcoma information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Gronchi A, Miah AB et al. Soft tissue and visceral sarcomas: ESMO-EURACAN-GENTURIS Clinical practice guidelines for diagnosis, treatment and follow-up. Annals of Oncology, 2021; 32, 11, 1348-1365 [accessed May 2022].

Casali PG, Blay JY et al. Gastrointestinal stromal tumours: ESMO-EURACAN-GENTURIS Clinical practice guidelines for diagnosis, treatment and follow-up. Annals of Oncology, 2022; 33,1, 20-33 [accessed May 2022].

Reviewers

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This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by senior medical editor Fiona Cowie, Consultant Clinical Oncologist.

Our cancer information has been awarded the PIF TICK. Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

Date reviewed

Reviewed: 01 August 2022 | Next review: 01 August 2025

This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

How we can help

Macmillan Support Line

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The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.

Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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